

Supplemental Material for “Topological Band Theory for Non-Hermitian Hamiltonians”

Huitao Shen,¹ Bo Zhen,^{1,2} and Liang Fu¹

¹*Department of Physics, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, USA*

²*Department of Physics and Astronomy, University of Pennsylvania, Philadelphia, Pennsylvania 19104, USA*

I. PROOF OF NON-ORTHOGONALITY OF LEFT AND RIGHT EIGENSTATES

Suppose the eigenvalues of H are non-degenerate. This means the Jordan normal form of H is diagonal: $H = PJP^{-1}$, where P is an invertible matrix and J is diagonal. First rewrite it as $HP = PJ$. Then the n -th column of P is a right eigenstate of H , corresponding to the eigenvalue E_n , denote as $|\psi_n^R\rangle$. Then rewrite $HP = PJ$ as $H^\dagger(P^{-1})^\dagger = (P^{-1})^\dagger J^*$. Now the m -th column of $(P^{-1})^\dagger$ is a left eigenstate of H , corresponding to the eigenvalue E_m^* , denoted as $|\psi_m^L\rangle$. Finally, we notice that $P^{-1}P = I$. This means $\langle\psi_m^L|\psi_n^R\rangle = \delta_{mn}$ under this normalization condition. Under a general normalization condition $\langle\psi_n^L|\psi_n^R\rangle \neq 0$, i.e., the left and right eigenstates corresponding to the same eigenvalue are non-orthogonal.

II. PROOF OF CHERN NUMBER EQUALITIES

Since the proof is for any given band, we suppress the band index n in the following. Our proof relies on the fact the Chern number is an obstruction to a global gauge of the wavefunction in the whole BZ [1], and the gauges of the left and right eigenstates are “locked” to each other because $\langle\psi_n^L|\psi_n^R\rangle \neq 0$.

We first prove $N^{\text{LR}} = N^{\text{RR}}$. It is convenient to introduce the Berry connection for a given band:

$$\mathbf{A}^{\alpha\beta}(\mathbf{k}) = \langle\psi^\alpha(\mathbf{k})|\nabla_{\mathbf{k}}\psi^\beta(\mathbf{k})\rangle. \quad (1)$$

Here $\alpha, \beta = \text{L/R}$.

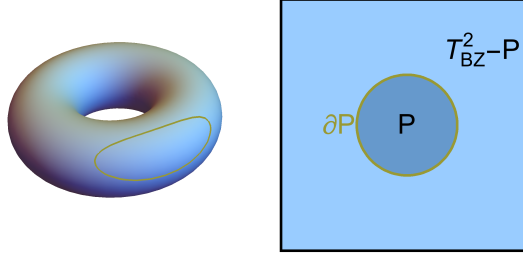


FIG. 1. (Color online) The schematic diagram of the patch P in the 2D BZ.

Consider a patch P in the BZ, whose boundary ∂P has circumference L . A schematic diagram is shown in Fig. 1. We first choose a local gauge I in P for the right eigenstate, denote as $|\psi_I^R\rangle$. We then choose another gauge II in $T_{\text{BZ}}^2 - P$, denote as $|\psi_{\text{II}}^R\rangle$. (We can always divide the BZ into more patches and follow the similar arguments below if two patches are not enough for well-defined local gauges.) For the case $\alpha = \beta = \text{R}$, the two gauges are related by the gauge transformation

$$|\psi_{\text{II}}^R(\mathbf{k})\rangle = e^{if(\mathbf{k})} |\psi_I^R(\mathbf{k})\rangle \quad (2)$$

on the boundary ∂P . For the case $\alpha = \text{L}$ and $\beta = \text{R}$, on the boundary

$$\begin{aligned} |\psi_{\text{II}}^R(\mathbf{k})\rangle &= r(\mathbf{k}) e^{if(\mathbf{k})} |\psi_I^R(\mathbf{k})\rangle, \\ |\psi_{\text{II}}^L(\mathbf{k})\rangle &= \frac{1}{r(\mathbf{k})} e^{if(\mathbf{k})} |\psi_I^L(\mathbf{k})\rangle. \end{aligned} \quad (3)$$

Both $r(\mathbf{k})$ and $f(\mathbf{k})$ are continuous real functions. Eq. (2) and Eq. (3) satisfy $\langle \psi^R(\mathbf{k}) | \psi^R(\mathbf{k}) \rangle = 1$ and $\langle \psi^L(\mathbf{k}) | \psi^R(\mathbf{k}) \rangle = 1$ respectively. Straightforward calculation gives the transformation law of the Berry connection

$$\mathbf{A}_{II}^{RR}(\mathbf{k}) = \mathbf{A}_I^{RR}(\mathbf{k}) + i \nabla_{\mathbf{k}} f(\mathbf{k}) \quad (4)$$

$$\mathbf{A}_{II}^{LR}(\mathbf{k}) = \mathbf{A}_I^{LR}(\mathbf{k}) + i \nabla_{\mathbf{k}} f(\mathbf{k}) + \frac{\nabla_{\mathbf{k}} r(\mathbf{k})}{r(\mathbf{k})}. \quad (5)$$

Berry curvature can be seen as the curl of the Berry connection: $\epsilon_{ij} B_{ij}^{\alpha\beta} = [\nabla \times \mathbf{A}]_z$, which is gauge independent due to $\nabla \times (\nabla f) = 0$. Stokes theorem implies

$$\begin{aligned} \int_{BZ} \epsilon_{ij} B_{ij}^{\alpha R} d^2 \mathbf{k} &= \oint_{\partial P} (\mathbf{A}_I^{\alpha R} - \mathbf{A}_{II}^{\alpha R}) \cdot d\mathbf{l} \\ &= i \oint_{\partial P} \nabla_{\mathbf{k}} f(\mathbf{k}) \cdot d\mathbf{l} = i(f(L) - f(0)), \end{aligned} \quad (6)$$

where $\alpha = L/R$. Notice that this number is quantized, pure imaginary, and is independent of the choice of α . There is no dependence of $r(\mathbf{k})$ because

$$\oint_{\partial P} \frac{\nabla_{\mathbf{k}} r(\mathbf{k})}{r(\mathbf{k})} \cdot d\mathbf{l} = \oint_{\partial P} \nabla_{\mathbf{k}} \log r(\mathbf{k}) \cdot d\mathbf{l} = \log \frac{r(L)}{r(0)} = 0. \quad (7)$$

Hence we have proved $N^{LR} = N^{RR}$. Physically, this is because the normalization condition $\langle \psi^\alpha(\mathbf{k}) | \psi^R(\mathbf{k}) \rangle = 1$ fixes the gauge of $|\psi^\alpha(\mathbf{k})\rangle$, and hence the Chern number is solely determined by the gauge of the right eigenstate.

With the same spirit, we can prove $N^{RL} = N^{LL}$. Notice $B^{LR}(\mathbf{k}) = -(B^{RL}(\mathbf{k}))^\dagger$. Since the integral of the Berry curvature is purely imaginary according to Eq. (6), we have $N^{LR} = N^{RL}$. Combine all the equalities, we finally finished the proof $N^{RR} = N^{LR} = N^{RL} = N^{LL}$.

We note that although the Chern numbers are the same, the gauge invariant Berry curvatures are locally different. In fact, the left-right Berry curvature is a complex number while the left-left and right-right Berry curvatures are purely imaginary.

We have verified all the results in this section numerically using the generalized Dirac model Eq. (4) in the main text.

III. VANISHING OF THE CHERN NUMBERS

Since we have proved $N^{RR} = N^{LR} = N^{RL} = N^{LL}$, we only consider N^{LR} in the following. According to Eq. (3) in the main text, the Chern numbers can vanish because

(a) The Berry curvature vanishes at all $\mathbf{k}$, i.e.,

$$\langle \partial_{k_x} \psi^L(\mathbf{k}) | \partial_{k_y} \psi^R(\mathbf{k}) \rangle = \langle \partial_{k_y} \psi^L(\mathbf{k}) | \partial_{k_x} \psi^R(\mathbf{k}) \rangle, \quad (8)$$

meaning $|\psi^L(\mathbf{k})\rangle = |\psi^R(\mathbf{k})\rangle^*$. This may result from $H(\mathbf{k}) = H(\mathbf{k})^T$, which is equivalent to a complex symmetric Hamiltonian. For 2×2 Hamiltonians $H = \mathbf{b} \cdot \boldsymbol{\sigma}$, this means $b_y = 0$. Only two Pauli matrices are allowed.

(b) The Berry curvature cancels in the whole BZ, i.e.

$$\begin{aligned} &\langle \partial_{k_x} \psi^L(\mathbf{k}) | \partial_{k_y} \psi^R(\mathbf{k}) \rangle - \langle \partial_{k_y} \psi^L(\mathbf{k}) | \partial_{k_x} \psi^R(\mathbf{k}) \rangle \\ &= \langle \partial_{k_y} \psi^L(-\mathbf{k}) | \partial_{k_x} \psi^R(-\mathbf{k}) \rangle - \langle \partial_{k_x} \psi^L(-\mathbf{k}) | \partial_{k_y} \psi^R(-\mathbf{k}) \rangle. \end{aligned} \quad (9)$$

This indicates $|\psi^L(\mathbf{k})\rangle = |\psi^R(-\mathbf{k})\rangle^*$, which may result from $H(\mathbf{k}) = H(-\mathbf{k})^T$.

The relation of these conditions with the symmetry of the Hamiltonian will be discussed in a separate paper.

IV. BULK-EDGE CORRESPONDENCE IN 2D

In this section, we first discuss the numerical solution of the domain wall problem (Eq. (5) in the main text) in detail. Then we show the numerical result of a similar problem on the square lattice to strengthen the discussion.

In order for the ansatz Eq. (6) in the main text to satisfy Eq. (5) in the main text, the following equations and conditions should be satisfied:

(i) Wavefunctions at both domains have the same energy:

$$\begin{aligned} E(k) &\equiv \sqrt{(m_1 + i\delta_1)^2 + (k + i\kappa_{1,y})^2 - (1/\lambda_+ - \kappa_{1,x})^2} \\ &= \sqrt{(m_2 + i\delta_2)^2 + (k + i\kappa_{2,y})^2 - (1/\lambda_- - \kappa_{2,x})^2}, \end{aligned} \quad (10)$$

(ii) Continuity of the wavefunction at $x = 0$:

$$\frac{(m_1 + i\delta_1) \pm E(k)}{(k + i\kappa_{1,y}) - (1/\lambda_+ - \kappa_{1,x})} = \frac{(m_2 + i\delta_2) \pm E(k)}{(k + i\kappa_{2,y}) - (1/\lambda_- - \kappa_{2,x})}, \quad (11)$$

(iii) Localized edge state

$$\text{Re}(1/\lambda_+) > 0, \text{Re}(1/\lambda_-) < 0. \quad (12)$$

$E(k)$ is generally complex and this makes an analytical solution hard to obtain. The imaginary part of $E(k)$ depends on κ_y and δ , and vanishes only under two circumstances: (a) When $\kappa_y = 0$. The reason is that κ_x only gives the mode propagating perpendicular to the domain wall a dissipation, since the boundary is along the y direction; (b) When the gain and loss provided by the two domains cancels delicately. For example, the imaginary part $\text{Im}(E(k=0))$ as a function of $\kappa_{1,y}$ and $\kappa_{2,y}$ is plotted in Fig. 2. It vanishes only when $\kappa_{1,y} + \kappa_{2,y} = 0$.

In the special case when one of the domains (say domain 2) is vacuum, we have $\delta_2 = \kappa_2 = 0$. The loss and gain balance could never be achieved, and the edge state (if exists) will always dissipate unless $\kappa_{1,y} = 0$.

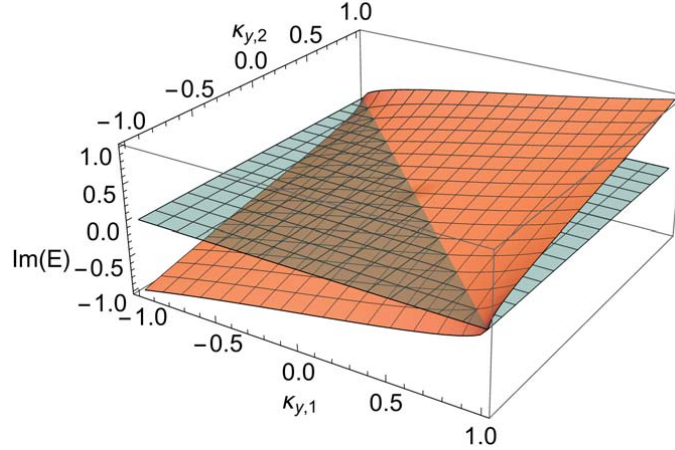


FIG. 2. (Color online) The imaginary part of the edge state energy $\text{Im}(E)$ at $k = 0$ as a function of $\kappa_{1,y}$ and $\kappa_{2,y}$. The transparent blue surface is the $\text{Im}(E) = 0$ equal energy surface. Here $m_1 = -m_2 < 0$ and $\kappa_{1,x} = \kappa_{2,x} = \delta_1 = \delta_2 = 0$. The energy unit is $|m_1| = |m_2|$.

To strengthen the discussion of the topological phase transition in the continuous Dirac model, here we consider a lattice model on a square lattice. The bulk Hamiltonian is given by

$$H(\mathbf{k}) = (t \sin k_x + i\kappa_x)\sigma_x + (t \sin k_y + i\kappa_y)\sigma_y + (t \cos k_x + t \cos k_y + m + i\delta)\sigma_z, \quad (13)$$

which is of the following form in real space

$$\begin{aligned} H = & -\frac{t}{2} \sum_{i,j} \left(c_{i,j+1,1}^\dagger c_{i,j,-1} - c_{i,j,1}^\dagger c_{i,j+1,-1} + \text{h.c.} \right) - \frac{t}{2} \sum_{i,j} \left(i c_{i+1,j,1}^\dagger c_{i,j,-1} - i c_{i,j,1}^\dagger c_{i+1,j,-1} + \text{h.c.} \right), \\ & + \sum_{i,j} \left[(i\kappa_x + \kappa_y) c_{i,j,1}^\dagger c_{i,j,-1} + (i\kappa_x - \kappa_y) c_{i,j,-1}^\dagger c_{i,j,1} \right] + (m + i\delta) \sum_{i,j,\sigma=\pm 1} \sigma c_{i,j,\sigma}^\dagger c_{i,j,\sigma} \\ & - \frac{t}{2} \sum_{i,j,\sigma=\pm 1} \sigma \left(c_{i+1,j,\sigma}^\dagger c_{i,j,\sigma} + c_{i,j+1,\sigma}^\dagger c_{i,j,\sigma} + \text{h.c.} \right), \end{aligned} \quad (14)$$

where $\sigma = \pm 1$ represents two orbitals. In the following, $t = 1$ is the energy unit. In the Hermitian limit $\kappa_x = \kappa_y = 0$, this model hosts a topological phase transition. $|m| < 2$ corresponds to the topologically nontrivial phase and $|m| > 2$

corresponds to the topological trivial phase. There are band degeneracies at $|m| = 2$ where four Dirac fermions are located on $\mathbf{k} = (0, 0), (0, \pi), (\pi, 0)$ and (π, π) .

In order to see the effect of edge states, we numerically compute the complex spectrum of this model under a cylinder geometry. The lattice is periodic along y direction, and has a finite number of n sites along x direction. In Fig. 3 we present the complex spectrum of a topologically nontrivial system with $n = 40$. In both of the panels, the bulk spectrum is separable. We can see clearly that two chiral edge states within the complex-energy “gap” connect the bulk bands. In the upper panel, the edge states do not dissipate. In the lower panel, the edge states have a constant imaginary energy due to the nonzero κ_y . These results are consistent with those of the domain wall problem perfectly discussed in the main text.

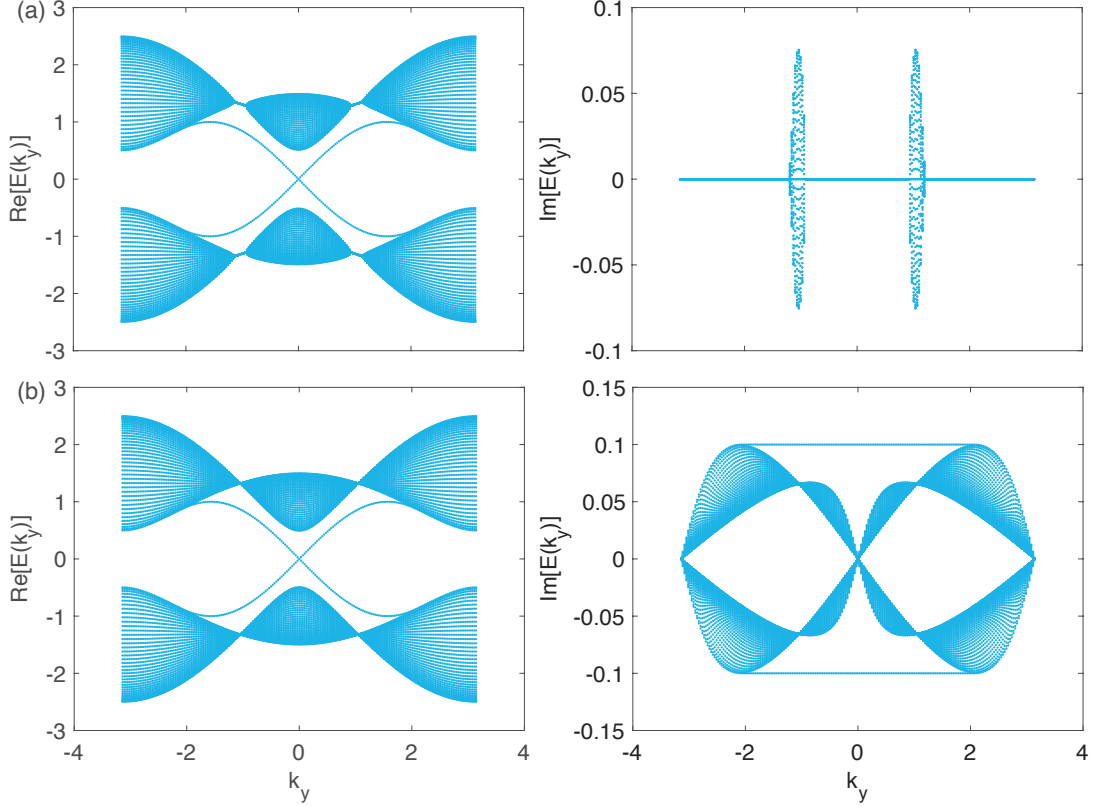


FIG. 3. (Color online) The complex energy spectrum of Eq. (13). The energy unit is the hopping strength t . The number of sites along x -direction is $n = 40$. The system is in the topologically nontrivial phase $m = -0.5, \delta = 0$. (a) $\kappa_x = 0.1, \kappa_y = 0$; (b) $\kappa_x = 0, \kappa_y = 0.1$.

V. NON-HERMITIAN LEVEL DEGENERACY REQUIRES TUNING 2 PARAMETERS

In this section, we review the well-established result that finding a level degeneracy in non-Hermitian Hamiltonians generically requires tuning 2 parameters [2].

It suffices to consider an 2×2 effective Hamiltonian H near the degeneracy point, obtained by projecting the generic Hamiltonian $\hat{H}$ to the subspace of these two levels. The most general form of H is

$$H = aI + (\mathbf{b}_0 + i\mathbf{b}_1) \cdot \boldsymbol{\sigma}, \quad (15)$$

where I is the identity matrix, $\boldsymbol{\sigma} \equiv (\sigma_x, \sigma_y, \sigma_z)$ denotes Pauli matrices, parameter a is complex, and $\mathbf{b}_i = (b_{i,x}, b_{i,y}, b_{i,z})$ with $i = 0, 1$ are real. The spectrum of H is

$$E_{\pm} = a \pm \sqrt{\mathbf{b}_0 \cdot \mathbf{b}_0 - \mathbf{b}_1 \cdot \mathbf{b}_1 + 2i\mathbf{b}_0 \cdot \mathbf{b}_1}. \quad (16)$$

Degeneracy of eigenvalues ($E_+ = E_-$) occurs when

$$\mathbf{b}_0 \cdot \mathbf{b}_0 = \mathbf{b}_1 \cdot \mathbf{b}_1, \text{ and } \mathbf{b}_0 \cdot \mathbf{b}_1 = 0. \quad (17)$$

The two equations above, which are the necessary and sufficient condition for level degeneracy, can be generically satisfied by tuning 2 real parameters in the Hamiltonian. In the Hermitian case when $\mathbf{b}_1 = 0$, these two equations reduce to $\mathbf{b}_0 = 0$, requiring to tune all 3 components of $\mathbf{b}_0$.

At a degeneracy point, Eq. (17) implies the two vectors $\mathbf{b}_0$ and $\mathbf{b}_1$ are orthogonal and of equal length. Hence, without loss of generality we choose $\mathbf{b}_0 = b(1, 0, 0)$ and $\mathbf{b}_1 = b(0, 1, 0)$. The resulting non-Hermitian Hamiltonian is

$$H_0 = aI + b\sigma_+. \quad (18)$$

This can be considered as the most general Hamiltonian at a non-Hermitian degeneracy. Importantly, for $b \neq 0$, the above 2×2 matrix has only one eigenstate $(1, 0)$. Such a non-Hermitian Hamiltonian is called “defective”. Only for the special case $b = 0$, H_0 is proportional to identity and has two degenerate and linearly-independent eigenstates. This result shows that the generic degeneracy points of non-Hermitian Hamiltonians are defective.

VI. MERGING TWO EXCEPTIONAL POINTS

In this section, we systematically study all four different scenarios when two exceptional points are moved together. Recall (Eq. (11) and (12) in the main text) that the most general Hamiltonian near the exceptional point could be written as $H(\mathbf{k}) = aI + \epsilon\sigma_+ + \sum_{i,j} k_i c_{ij} \sigma_j$, with the dispersion

$$E_{\pm}(\mathbf{k}) = a \pm \sqrt{c_x k_x + c_y k_y}, \quad (19)$$

where $c_x = 2\epsilon(c_{xx} + ic_{xy})$ and $c_y = 2\epsilon(c_{yx} + ic_{yy})$.

The first two scenarios involve moving together two exceptional points with opposite charge, which we have already encountered when discussing the topological phase transitions.

(a) It is possible that $c_x = c_y = 0$ in Eq. (19) and there is no phase accumulation in the quadratic terms. This is actually the case when two exceptional points merge into the familiar Dirac point in the Hermitian Hamiltonians. The Hamiltonian

$$H = (k_x + i\delta)\sigma_x + k_y\sigma_y, \quad (20)$$

hosts two exceptional points at $\mathbf{k} = (0, \pm\delta)$ with vorticity $\pm 1/2$ respectively. As $\delta \rightarrow 0$, the exceptional points annihilate into the Dirac point, where there are neither interchange of the eigenvalues, nor the Hamiltonian is defective.

(b) There is another case when c_x and c_y in Eq. (19) have the same argument $\text{Im}(c_y/c_x) = 0$ so that there is no net phase accumulation when the degeneracy is encircled. With a proper rotation of the momentum to k'_x and k'_y , the dispersion must be proportional to $\sqrt{k'_x}$ at one direction and to k'_y in another direction. Based on the dispersion profile, we call this degeneracy “hybrid point”. We demonstrate hybrid point with the Hamiltonian

$$H = (k_x + i\delta)\sigma_x + k_y\sigma_y + m\sigma_z, \quad |\delta| \geq |m| > 0. \quad (21)$$

It has two exceptional points at $\mathbf{k} = (0, \pm\sqrt{\delta^2 - m^2})$ with opposite charges. When $|\delta| = |m|$, the two exceptional points merge at $\mathbf{k} = 0$ and the spectrum becomes $E_{\pm}(\mathbf{k}) = \pm\sqrt{k_x^2 + k_y^2 + 2ik_x m}$ (Fig. 4). It is obvious there is no interchange of the eigenvalues, although the Hamiltonian at this point is still defective.

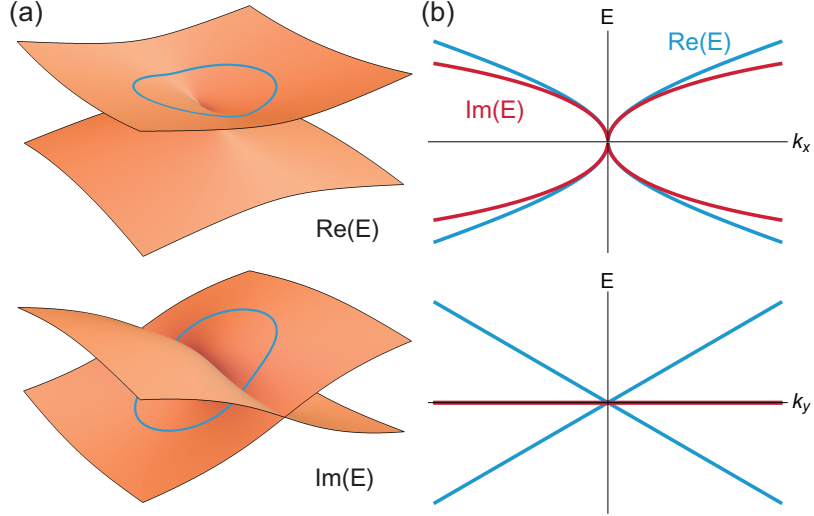


FIG. 4. (Color online) (a) The dispersion near a hybrid point. The Hamiltonian is Eq. (21) for $m = \delta = 1$. The blue loop is $k = 1$. Clearly there is no eigenvalue switching. (b) Same dispersion along $k_y = 0$ and $k_x = 0$, when the dispersion is proportional to $\sqrt{|k_x|}$ and $|k_y|$ respectively. This anisotropy is why the degeneracy is called “hybrid point”.

Moving together two exceptional points with the same charge gives the remaining two scenarios.

(c) The degeneracy is a “double exceptional point” with vorticity ± 1 , on which the Hamiltonian is defective. This can be seen from

$$H = (k_x + isk_y)\sigma_z + (\delta + \lambda)\sigma_x + i\delta\sigma_y, \quad \delta \neq 0. \quad (22)$$

Here $s = \pm 1$. There are two vorticity $s/2$ exceptional points at $\mathbf{k} = (0, \pm\sqrt{\lambda(\lambda + 2\delta)})$. As $\lambda \rightarrow 0$, $\mathbf{k} = 0$ is a double exceptional point.

(d) The Hamiltonian is not necessarily defective when the vorticity is an integer. Consider the previous Hamiltonian Eq. (22) with $\delta = 0$. The Hamiltonian at $\lambda \rightarrow 0$ is now just a zero matrix, which is of course not defective. Although there is no interchanging of the eigenstates, the eigenvalues have a integer phase winding like a vortex. We thus name this point “vortex point”.

TABLE I. Four types of degeneracy in non-Hermitian Hamiltonians with their properties.

Degeneracy	Vorticity	Defectiveness
Exceptional point	Half-integer	Defective
Hybrid point	Zero	Defective
Dirac point	Zero	Non-defective
Vortex point	Nonzero integer	Non-defective

All the four types of degeneracy are summarized in the Table. I. The non-zero winding of the eigenvalues is not necessarily related to the defectiveness of the Hamiltonian. Except the exceptional point, all three other types of degeneracy need fine-tuning parameters and any perturbation will make them into two exceptional points in 2D.

-
- [1] M. Kohmoto, Ann. Phys. **160**, 343 (1985).
[2] M. Berry, Czechoslov. J. Phys. **54**, 1039 (2004).